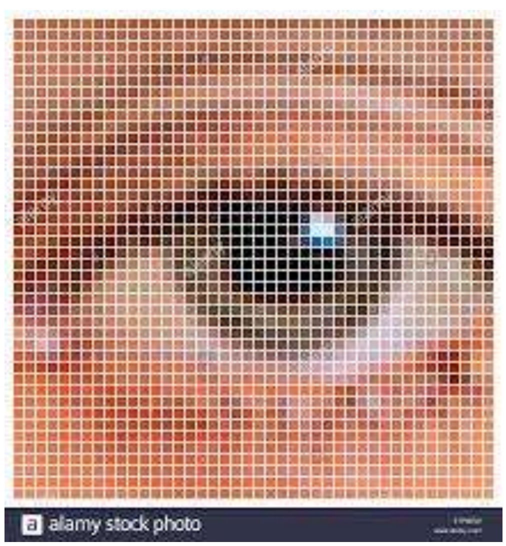
Concept of Pixel

Pixel

- Pixel is the smallest element of an image on a computer display, whether they are LCD or CRT monitors.
- A screen is made up of a matrix of thousands or millions of pixels.
- A pixel is represented with a dot or a square on a computer screen.
- Pixel cannot be seen as they are very small which result in a smooth and clear image.
- Each pixel has a value.
- It can have only one color at a time.
- Colour of a pixel is determined by the number of bits which is used to represent it.
- A resolution of a computer screen depends upon graphics card and display monitor, the quantity, size and color combination of pixels.



Relationship with CCD array:

When an image has zoomed in, the surface of CCD will look like filled dots. These dots are light receptor called photodiode.

CCD sizes are described using Terms like 2 million pixels (megapixel) and 4 million pixels (megapixel). The more the count of pixel, more detailed image is generated. To get a clear and image smooth, CCD and image size is increased.

<u>Calculation of the total number of pixels:</u>

Let the number of rows=300 & columns=200
Total number of pixels= 300 X 200
= 50000

Gray level

The value of the minimum gray level is 0. The gray level depends on the depth of the image.

For example: In an 8-bit image, gray level is 0 to 255.

For a binary image, a pixel can only take value 0 or 1.

In color image, it can choose values between 0 and 255 in different channel.

The formula for calculating gray level in a color image is as shown below:

Grey level = 0.299 * red component + 0.587 * green component + 0.114 * blue component

<u>Pixel value (0):</u>

As we know that each pixel has a unique value. 0 is a unique value that means the absence of light. It means that 0 is used to denote dark.

For example:

We have a matrix of 3X3 of an image, and each pixel is of value as shown below:

0	0	0
0	0	0
0	0	0

Total number of pixel= Number of rows × Number of column

$$= 3 \times 3$$

$$= 9$$

It means the image formed is of 9 pixels which are black.

The image would be something as below: